

Carbon price pass-through in Alberta's electricity market

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Carbon Pricing's Effect on Electricity Markets

- Our key question is how carbon prices impact firm offer behaviour in electricity markets? Since demand is nearly perfectly inelastic in the short run, these offer changes translate to price changes.
- Electricity markets, at least in certain jurisdictions, are ideal environments in which to study responses to carbon prices
- A series of policy changes in Alberta have altered both carbon prices and output-based credit allocations differently for different facilities over time, creating a policy experiment
- Rich, hourly electricity market data allows us to identify facility- and firm-level decisions with respect to carbon price pass through

Literature on Carbon Price Pass-Through

- Fabra and Reguant (2014) shows how firms respond to emissions credit prices argues that lump-sum credit allocations were not distortionary in the short run
- Brown et al. (2018) have looked at the dynamics of such a market in a simplified, simulation model

Other Related Literature

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Alberta's Power Market is an Ideal Laboratory

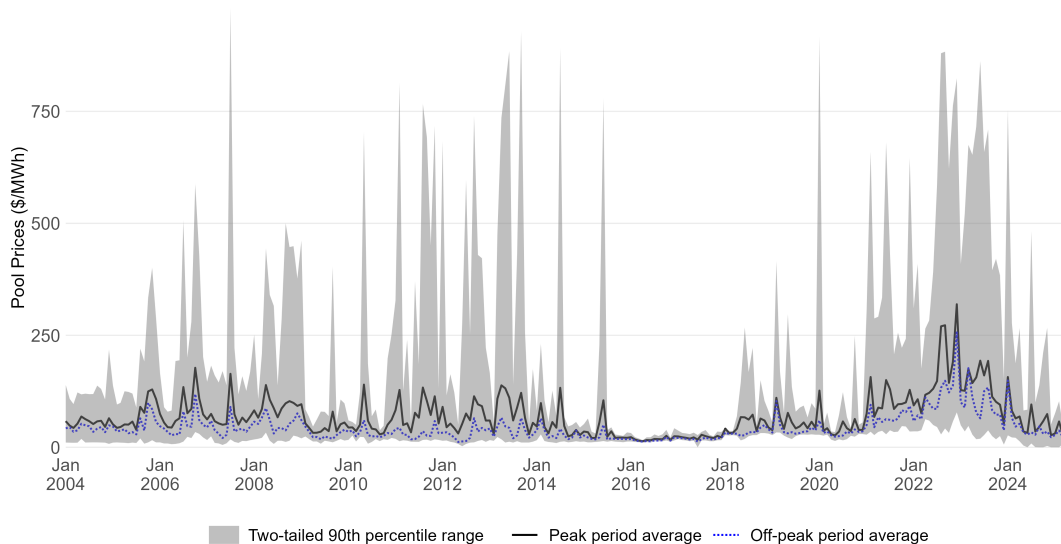
- Alberta's power market is relatively small, isolated, and features a mix of generation technologies
- Significant hourly variability in intertie capacity helps us with identification
- Alberta has minimal intertie capacity: WECC (1325 MW export and 1500 MW import path-rating) and Saskatchewan (150 MW path-rating)
- Peak internal load for 2018 was 11,697 MW, minimum internal load was 7,819 MW. Sustained low prices prevailed from 2015 through 2018.
- Supply mix (2018) was 45% coal, 21% net-to-grid from combined heat and power plants, 19% natural gas, 3% hydro, 6% wind, 1% other sources including solar and 5% imports.

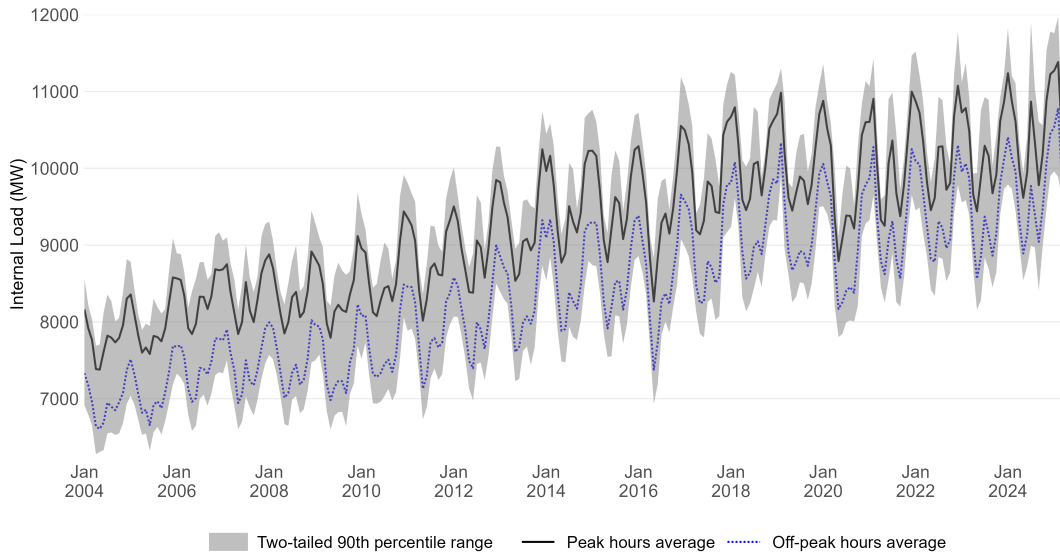
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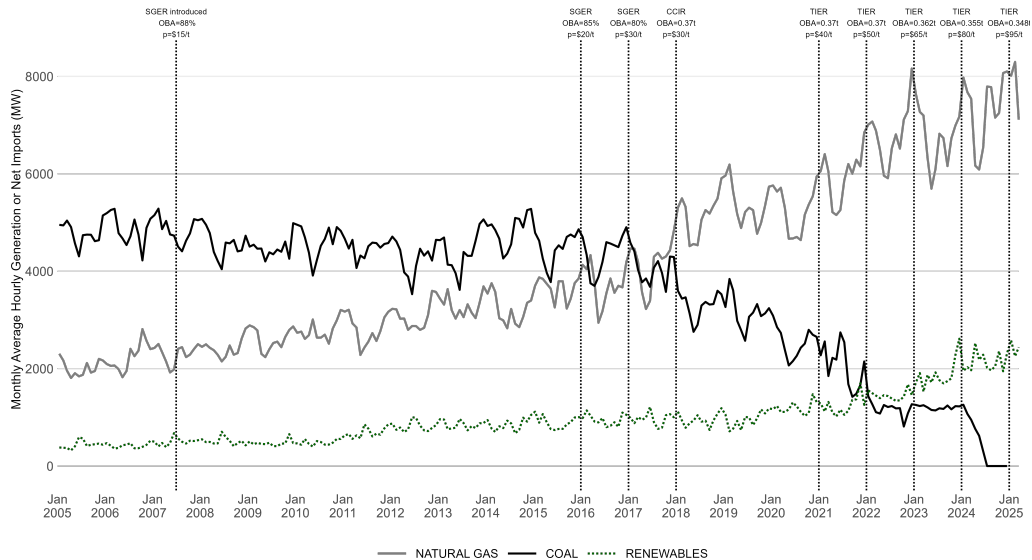
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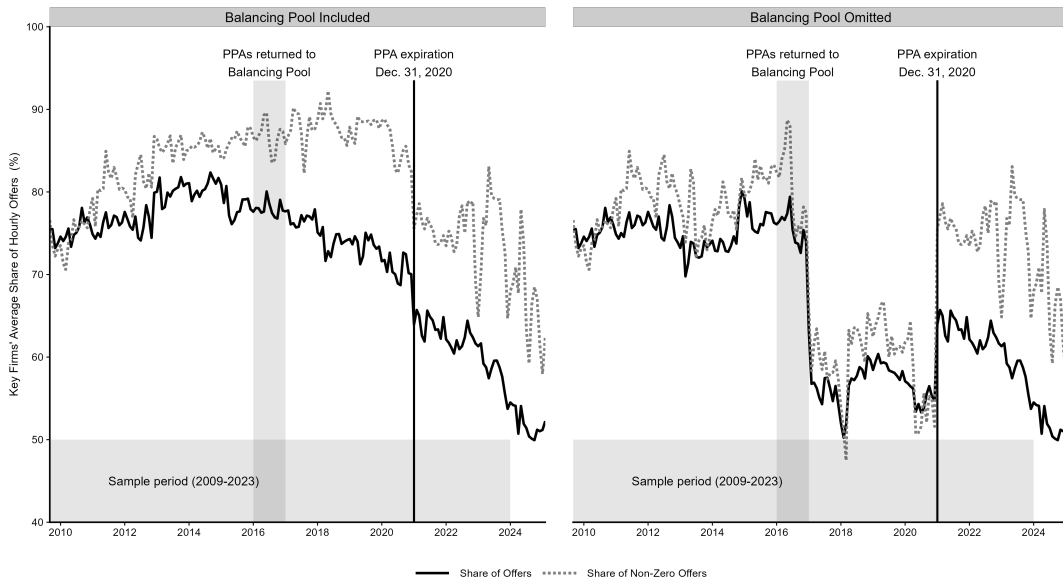
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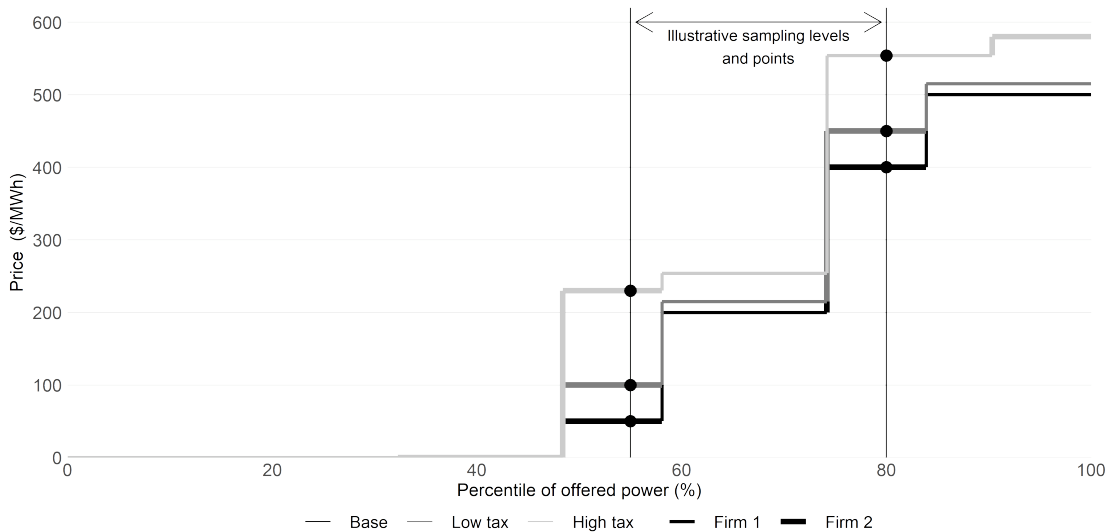
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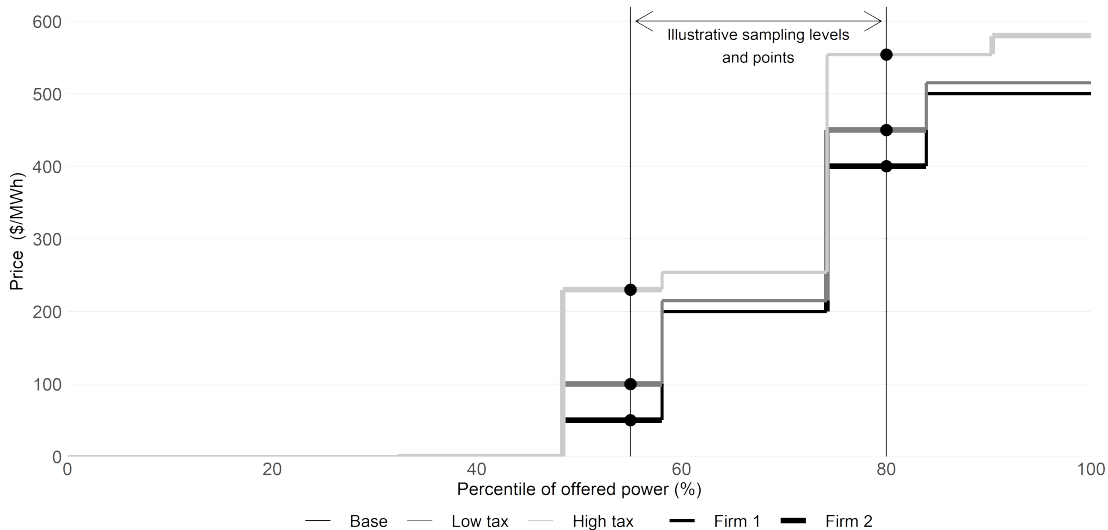




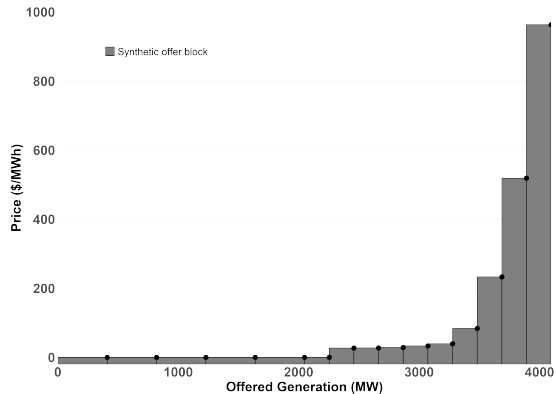
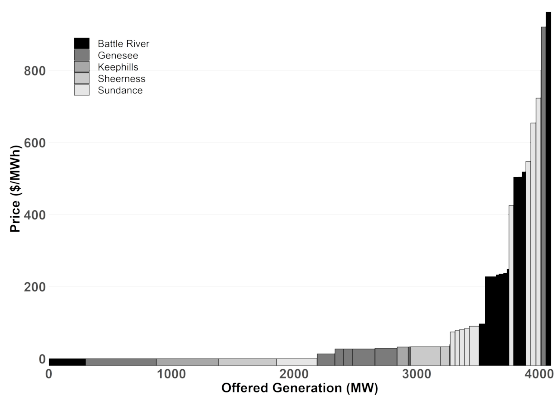




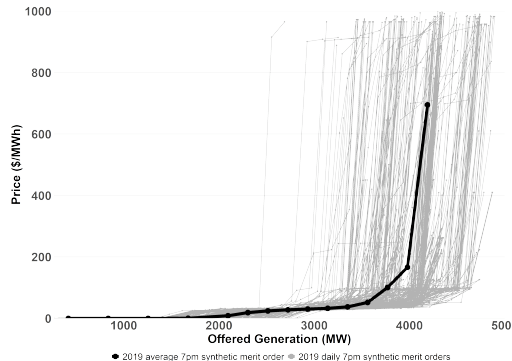
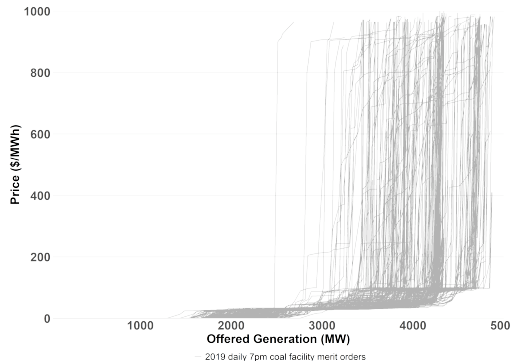




Merit Order Sampling



Merit Order Sampling with Coal Offers

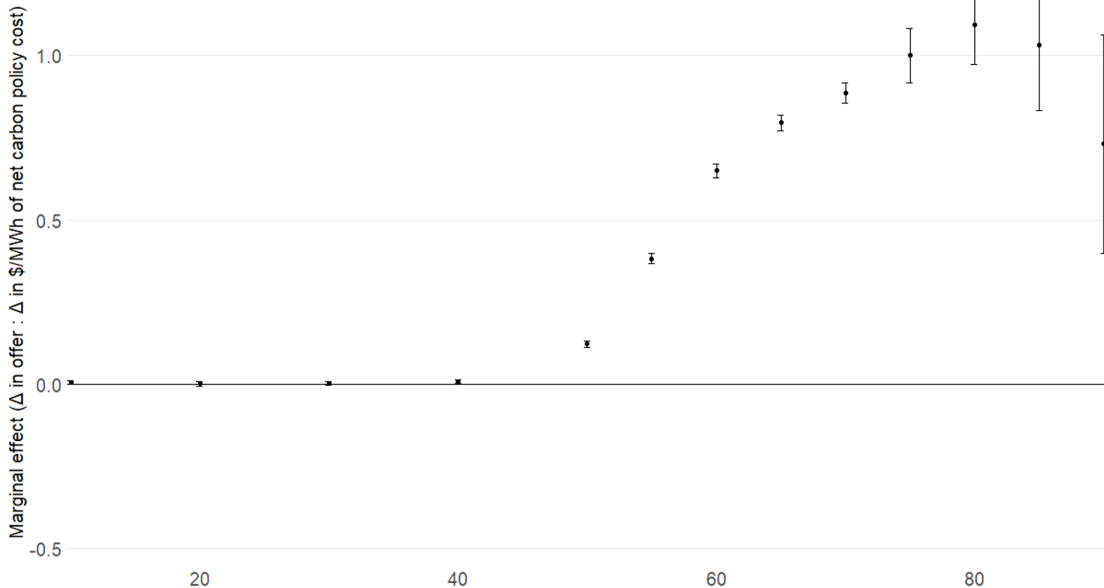


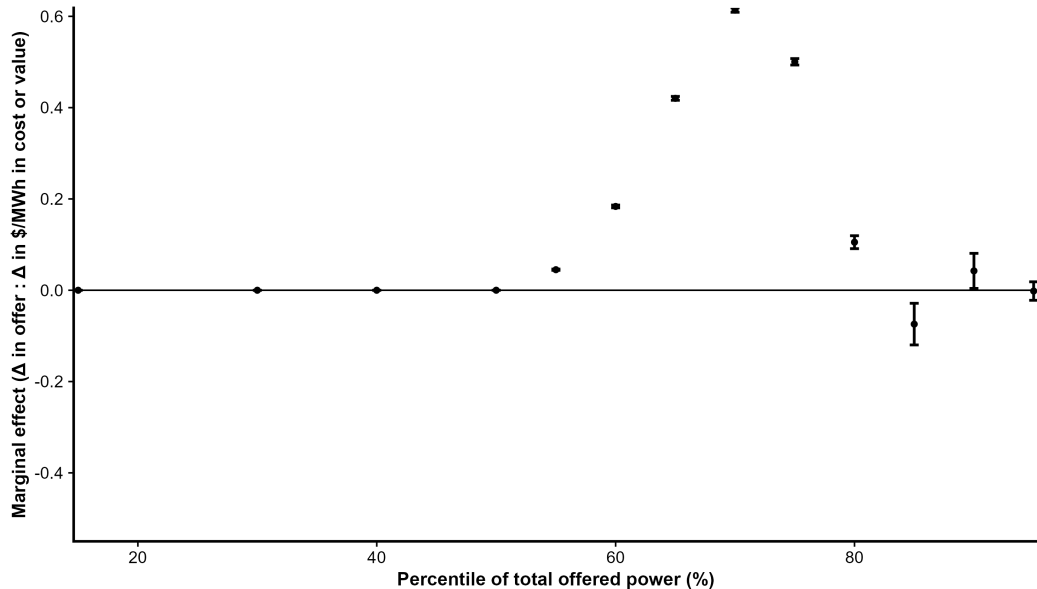
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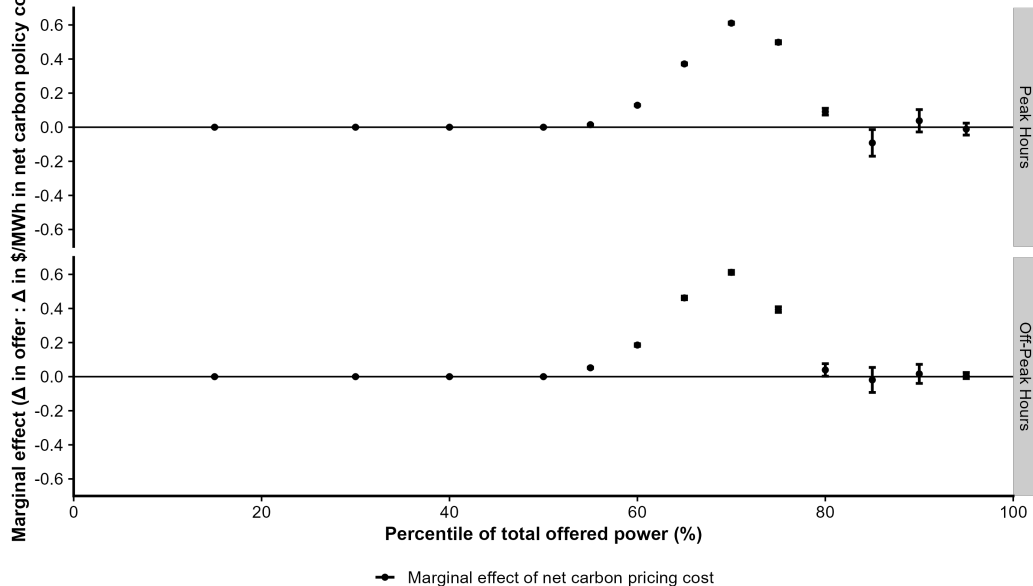
The variation in carbon pricing over time, as well as variation in which firms are participating in the market, and which facilities' offers are present at a sampling point in the merit order at different times all allow us to identify the impact of carbon costs on offer values. For each sampled point in the synthetic merit order, we identify facility (F) and market conditions (M), as well as carbon policy compliance costs, which we use to explain changes in the level offer levels over time. The regression equation is given by:

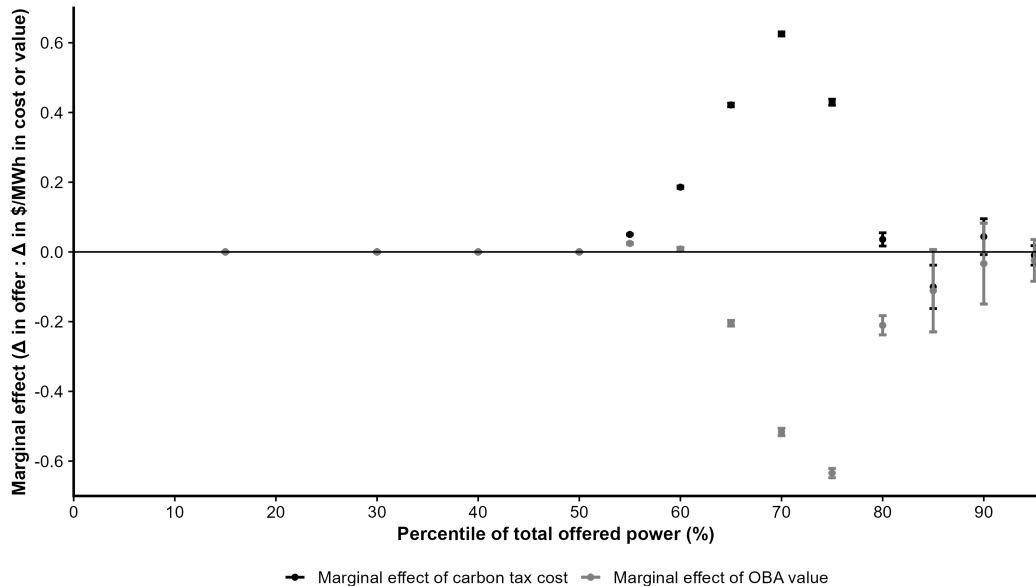
$$\underbrace{P_{j,t}}_{\text{Offer value at percentile } j \text{ at time } t} = \underbrace{\alpha_j}_{\text{Constant for percentile } j} + \underbrace{\beta_j M_t}_{\text{Market, policy, and climate variables at time } t} + \underbrace{\delta_j F_{j,t}}_{\text{Facility characteristics at percentile } j \text{ at time } t} + \underbrace{\zeta_j t_{j,t}}_{\text{Carbon cost at percentile } j \text{ at time } t} + \underbrace{\kappa_j O_{j,t}}_{\text{OBA value at percentile } j \text{ at time } t} + \underbrace{\epsilon_{j,t}}_{\text{Error term}}$$

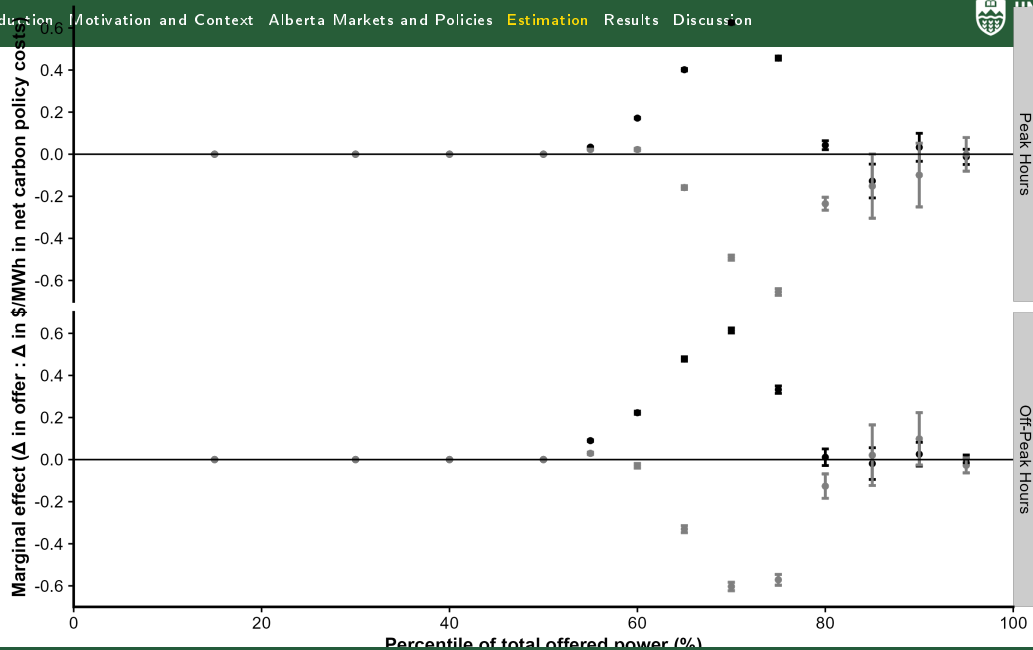
(1)

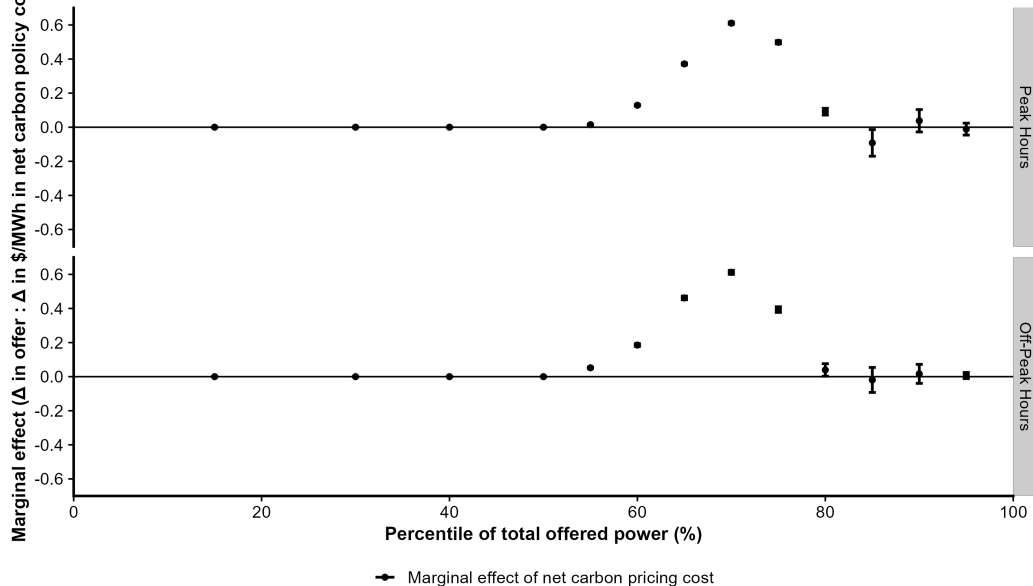
Marginal effect (Δ in offer : Δ in \$/MWh of net carbon policy cost)

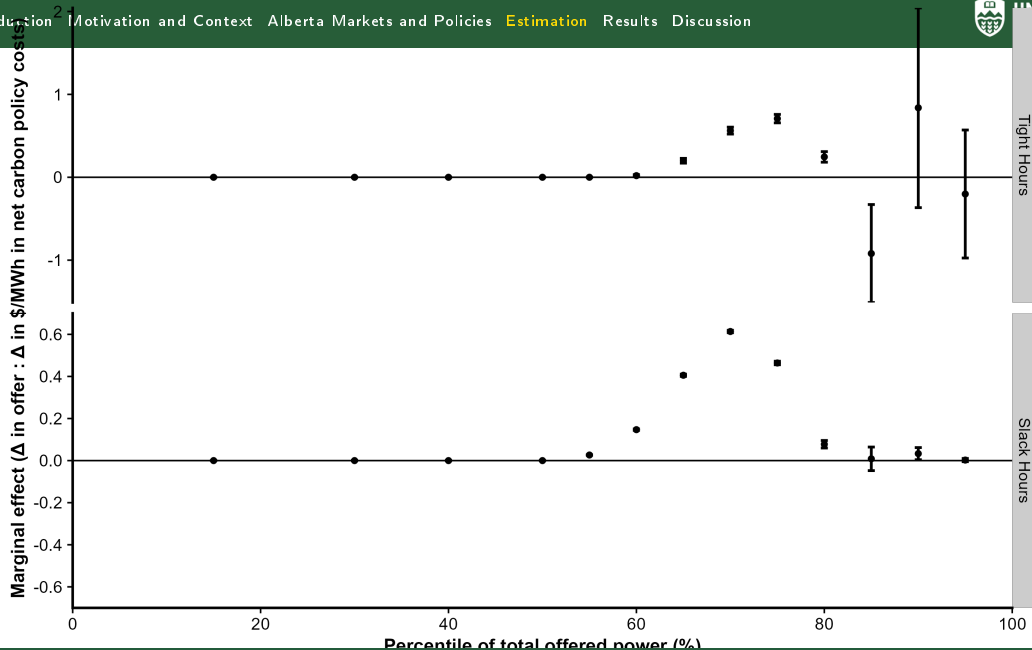












Marginal effect (Δ in offer : Δ in \$/MWh of carbon policy cost)

1.0
0.5
0.0
-0.5
-1.0
-1.5
-2.0

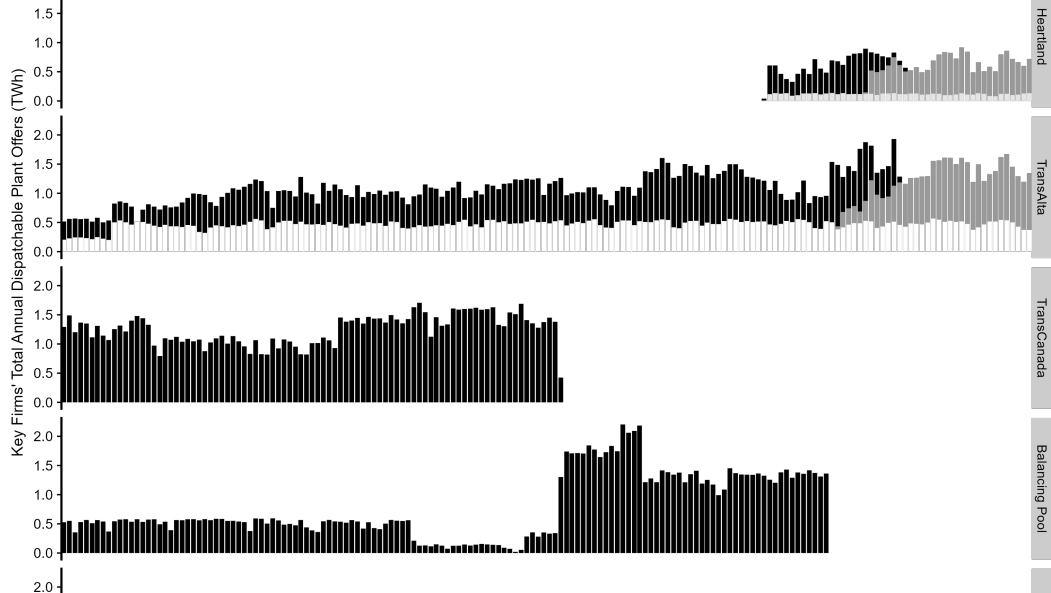
Gas-Fired
Steam

1
0
-1
-2
-3
-4

Natural Gas
Combined Cycle

4
2
0

Natural Gas
Simple Cycle





Facility	Plant Owner	Unit ID	Capacity (MW)	End of PPA Term	Returned to Balancing Pool	Conversion to gas-fired steam or retirement
Battle River	ATCO	BR3	147	Dec 31, 2013	NA	NA
		BR4	147	Dec 31, 2013	NA	Mar 8, 2022
		BR5	368	Dec 31, 2020	Jan 1, 2016	Nov 19, 2021
Genesee	Capital Power	GN1	381	Dec 31, 2020	NA	May 1, 2024*
		GN2	381	Dec 31, 2020	NA	July 1, 2024*
		GN3	466	NA	NA	July 12, 2024*
HR Milner	ATCO	HRM	144	Jan 01, 2013	NA	May 8, 2020
Keephills	TransAlta	KH1	383	Dec 31, 2020	May 5, 2016	Jan 11, 2022
		KH2	383	Dec 31, 2020	May 5, 2016	July 21, 2021
		KH3	466	NA	NA	Jan 11, 2022*
Sheerness	ATCO (50%)	SH1	378	Dec 31, 2020	Mar 7, 2016	July 30, 2021
	TransAlta (50%)	SH2	378	Dec 31, 2020	Mar 7, 2016	July 30, 2021
Sundance A	TransAlta	SD1	280	Dec 31, 2020	Mar 7, 2016	Jan 1, 2018*
		SD2	280	Dec 31, 2020	Mar 7, 2016	July 31, 2018*
Sundance B	TransAlta	SD3	353	Dec 31, 2020	Mar 7, 2016	July 31, 2020*
		SD4	353	Dec 31, 2020	Mar 7, 2016	Apr 1, 2022*
Sundance C	TransAlta	SD5	353	Dec 31, 2020	Mar 24, 2016	Nov 1, 2021*
		SD6	357	Dec 31, 2020	Mar 7, 2016	Feb 1, 2021

Note: * denotes a retirement as opposed to a conversion. Supercritical coal units at Keephills (KH3, 466MW, converted to gas Jan 11, 2022) and Genesee (GN3, 466 MW, converted to gas July 12, 2024) were never part of the PPA system, as they were built as merchant generators. A more extensive re-power of the Genesee 1 and 2 units saw them reclassified as natural gas combined-cycle plants after the retirements listed above.

Discussion

- Alberta's energy-only power market provides an ideal environment to study carbon price pass-through
- Alberta policy changes alter both carbon prices and output-based credit allocations provide facility-specific treatment
- Preliminary results show that policy changes in 2016, 2017, and 2018 saw significant changes in offers from coal and natural gas combined cycle plants
- Framework allows us to examine changes in firm, offer control, fuel or other dimensions as well as market power implications
- Machine-learning-based estimation of synthetic offer curves due to market conditions still to come
- Glass half-full: impact of new policies and other proposed changes might provide an additional treatment.

Contact info

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